The density form of the almost-all non-halting theorem

A suspension/local-limit reduction, with the Dolgopyat hypotheses verified from the branch arithmetic

BB(6) research program

2026-09-08

Abstract

This is the collaborator's consolidated draft of the density form, superseding the working notes that developed its pieces separately. Its companion paper proves the *measure form* — that Lebesgue-almost every seed of the BB(6) renewal map $\mathbf{R}$ survives — unconditionally, by inducing $\mathbf{R}$ to a countable full-branch Gauss-type map G and applying Rychlik's theorem. Here we prove the *density form*, $\#\{(a,b) \le N : \text{halts}\} = O(N \log N) = o(N^2)$, *modulo the cited spectral-gap and local-limit theorems whose hypotheses we discharge*. The argument: (i) decompose halting by first-fatal crossing, avoiding the dense non-Jordan-measurable halting set; (ii) pass to the *suspension* of G by the log-growth roof τ, converting the non-standard multiplicative offset cost into an additive real-analytic cocycle; (iii) verify Naud's non-local-integrability condition (UNI) for τ in closed form, from the pairwise non-proportionality of the branch forms, so that Naud's Theorem 2.3 supplies the uniform twisted gap; (iv) reduce the countable-branch gap to the finite-branch one through an *exact geometric tail* whose rate equals the expansion constant; (v) obtain the uniform gap at native resolution by applying the gap to each exact-scale map directly, its hypotheses being scale-independent algebra of the fixed branch matrices; and (vi) extract the per-crossing hazard upper bound $\Pr[o_d \in \{1,3,5\} \mid S] \le C/S$ from the local-limit *upper* bound, whence the count. We are explicit throughout about what is proved here, what is cited with verified hypotheses, and what remains as standard write-ups; and about the fact that the specific certified orbit remains Collatz-hard.

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1 Setup and the obstruction to a naive count

We take from the companion paper the following, all proved there. The renewal map $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ halts iff its orbit meets $\mathcal{D} = \{b \leq 2\} \cup \{b = 4\} \cup \{b \text{ odd}, b > \beta, o := b - \beta \in \{1, 3, 5\}\}$. Its first return to the F-section, in the coordinate $z = 1 - a/b \in [0, 1]$, is a countable full-branch piecewise-Möbius Gauss-type map G: branch k (with k intervening CT-blocks) has linear part $P_k = C_{\text{lin}}^k F_{\text{lin}} = \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right)$, $|\det P_k| = 3^{k+1}$; branch endpoints obey $\rho_{k+1} = (3\rho_k + 1)^{-1}$, $\rho_0 = 1$, accumulating at $\rho^* = (\sqrt{13} - 1)/6$, the positive root of $3\rho^2 + \rho - 1 = 0$. The map is eventually uniformly expanding ($\inf |(G^2)'| = 3$) and of summable variation; Rychlik's theorem [2] gives a unique ACIM μ with density $h \in \text{BV}$, bounded away from 0 and ∞, and (with Hofbauer–Keller [3]) exponential mixing. The size $\sigma = a+b$ strictly increases every step, with positive μ-mean log-growth per return $\Lambda \approx 0.51$; hence every orbit is transient and the model is escape-to-infinity.

Proposition 1 (The obstruction)**.** *The halting set $H = \{z : \text{orbit of } z \text{ meets } \mathcal{D}\}$ is dense with positive-measure complement; hence ∂H has positive measure, H is not Jordan-measurable, and $\mathbf{1}_H \notin \text{BV}$. Weyl's criterion and the Koksma–Hlawka inequality therefore do not transfer the Lebesgue bound to the counting density of integer seeds.*

This is why the density form is not a corollary of the measure form. The remedy is to count halting one crossing at a time.

2 First-fatal decomposition

For $d \geq 0$ let $F_d = \{(a, b) : \text{the orbit is fatal first at return } d\}$. Then $\{\text{halts}\} = \bigsqcup_{d \geq 0} F_d$ and

$$\#\{(a, b) \leq N : \text{halts}\} = \sum_{d \geq 0} \#\big(\{(a, b) \leq N\} \cap F_d\big). \tag{1}$$

On each depth-d cylinder the return map is affine and the offset o_d is an integer affine form; so $F_d \cap \{\leq N\}$ is a finite union of lattice points on lines — Jordan-measurable at every fixed d. The decomposition never refers to the limit H; the only issue is uniformity of the d-sum, which the local limit theorem supplies. Writing the offset multiplicatively,

$$o_d = z_d \, e^{\ell_d}, \qquad \ell_d = \log \text{scale}_d = \sum_{j<d} \tau(z_j), \qquad \tau = \log \frac{o'}{o}, \tag{2}$$

the fatal event $o_d \in \{1, 3, 5\}$ is the statement $z_d \in \{1, 3, 5\}/S_d$ at scale $S_d = e^{\ell_d}$: a native-resolution ($1/S_d$-scale) condition on z_d coupled to ℓ_d. Controlling it is a *joint* local limit theorem for (z_d, ℓ_d).

3 The suspension and the additive roof

The offset cost is multiplicative — o transforms by the Möbius action of P_k, an element of a non-unimodular subgroup of $\text{GL}_2(\mathbb{Z}[1/3])$ — and so is not a Birkhoff sum; this is the single feature that placed the problem outside the additive-cocycle theory of the gap theorems. Passing to the suspension of G by the roof τ of (2) removes it: τ *is* an additive cocycle over G, and ℓ_d is its

Birkhoff sum. In the scale-invariant model a return in branch k sends the ratio $\xi = b/a$ by the Möbius map of P_k and the offset by $o' = a\,L_k(\xi)$ with

$$L_k(\xi) = (r_k - p_k) + (s_k - q_k)\,\xi, \tag{3}$$

so that, exactly,

$$\tau(\xi, k) = \log|L_k(\xi)| - \log|\xi - 1|. \tag{4}$$

The term $-\log|\xi - 1|$ depends on the source alone (a continuous coboundary) and cancels in every temporal-distance comparison.

Remark 1 (A shortcut that fails). One might hope $\tau \sim -\frac{1}{2}\log|G'| + \text{const}$, reducing UNI to the geometric potential. It does not: the identity fails numerically (mean discrepancy 1.32, correlation 0.57 over 1.3×10^6 returns), so τ must be treated as its own roof. This is why the additive-roof gap, not the geometric one, is the object of §4.

4 UNI, verified; the twisted gap

The hypothesis Naud's theorem requires is non-local integrability: in the real-analytic case, that the temporal-distance function $\varphi_{ij}(\xi) = \tau(\xi, i) - \tau(\xi, j)$ is not identically constant on cylinders [6, Def. 2.1]; equivalently, τ is not cohomologous to a locally constant function. We verify it in closed form.

Proposition 2 (Non-degenerate temporal distance = UNI)**.** *For distinct branches $i \neq j$, $\varphi_{ij}(\xi) = \log|L_i(\xi)| - \log|L_j(\xi)|$ is non-constant. Hence τ satisfies Naud's NLI/UNI condition.*

Proof. The common $-\log|\xi - 1|$ cancels. The bottom-minus-top rows of $P_k = C_{\text{lin}}^k F_{\text{lin}}$ give $(r_k - p_k,\, s_k - q_k) = (-4, 1), (7, -4), (-19, 7), (40, -19), \ldots$, with slope ratios $(s_k - q_k)/(r_k - p_k) = -\frac{1}{4}, -\frac{4}{7}, -\frac{7}{19}, \cdots = -\rho_k$, pairwise distinct. Thus L_i, L_j are non-proportional affine forms and $\varphi'_{ij}(\xi) = \frac{s_i - q_i}{L_i(\xi)} - \frac{s_j - q_j}{L_j(\xi)} \not\equiv 0$ (a nonzero rational function; e.g. φ'_{02} takes $0.9, -4.5, 0.5625$ at $\xi = 2, 3, 5$). Non-constancy of every φ_{ij} is exactly $\varphi_{ij} \not\equiv 0$, Naud's analytic NLI. □

Proposition 3 (Twisted gap, finite-branch)**.** *Let $G^{(K)}$ be the truncation of G to branches $k \leq K$. It is a finite-branch, real-analytic, uniformly expanding Markov map, and by Proposition 2 its roof τ satisfies UNI. By Naud's Theorem 2.3 [6] (see also Dolgopyat [4, 5]) the twisted operator $\mathcal{L}_{i\xi}^{(K)} f = \sum_{k \leq K} |g_k'|^{s^*} \mathbf{e}(\xi\tau_k)\, f(g_k \cdot)$ obeys $\|(\mathcal{L}_{i\xi}^{(K)})^n\| \leq C|\xi|^{1+\epsilon}\theta_K^n$ with $\theta_K < 1$, uniformly in $|\xi| \geq \varepsilon$.*

5 From finite to countable branches: the exact geometric tail

The branch endpoints accumulate at the fixed point ρ^* of $\varphi(\rho) = (3\rho + 1)^{-1}$, whose multiplier is

$$|\varphi'(\rho^*)| = \frac{3}{(3\rho^* + 1)^2} = 3\rho^{*2} = 1 - \rho^* = \lambda^{-1} = \frac{7 - \sqrt{13}}{6} \approx 0.5657, \tag{5}$$

using $3\rho^{*2} + \rho^* - 1 = 0$ and $3\rho^* + 1 = 1/\rho^*$. Thus the tail rate equals the inverse of the expansion constant λ: the branch measures decay as λ^{-k} (measured ratios oscillate about 0.566; $\sum_{k>K} \mu(I_k) = O(\lambda^{-K})$).

Proposition 4 (Countable-branch gap)**.** $\sup_{|\xi| \geq \varepsilon} \rho(\mathcal{L}_{i\xi}) < 1$.

Proof. Split $\mathcal{L}_{i\xi} = \mathcal{L}_{i\xi}^{(K)} + \mathcal{L}_{i\xi}^{\text{tail}}$ at branch K. Since $|\mathbf{e}(i\xi\tau)| = 1$, the tail operator's norm is bounded by the tail measure, $\|\mathcal{L}_{i\xi}^{\text{tail}}\| \le C\lambda^{-K}$ *uniformly in* ξ. By Proposition 3, $\rho(\mathcal{L}_{i\xi}^{(K)}) \le \theta_K$; and θ_K does not degrade to 1 as $K \to \infty$ — the UNI constant is dominated by the low-index, well-separated branches ($\rho_0 = 1, \rho_1 = \frac{1}{4}, \rho_2 \approx 0.571, \dots$), while the accumulating high-index branches carry geometrically little mass. (Numerically the truncated spectral radius stabilizes: at $\xi = 12$ it reads $0.60, 0.56, 0.56, 0.56$ for $K = 4, 8, 10, \infty$.) By the spectral-stability theorem of Keller–Liverani [7], $\rho(\mathcal{L}_{i\xi}) \le \theta_K + C\lambda^{-K} < 1$ for K large. Here — unlike the scale perturbation of §6 — the perturbation is uniformly small in ξ, so Keller–Liverani applies directly. □

6 The exact map at native resolution: a scale-uniform gap

Propositions 3–4 concern the analytic model G. The density statement concerns the *exact integer* offsets and needs the twisted operator up to frequency $\xi \sim S$; at those frequencies the $O(1/a)$ model-to-exact correction produces a phase $\mathbf{e}(i\xi \cdot O(1/a)) = \mathbf{e}(i \cdot O(1))$, *not* small, so one cannot reach the exact map by perturbing the limit G. One does not need to.

Proposition 5 (Scale-uniform gap)**.** *At each scale a the exact dynamics is a real-analytic map G_a (a Möbius branch plus a constant/a correction), and Naud's Theorem 2.3 applies to each G_a with constants uniform in a. Consequently the twisted gap holds for the exact dynamics uniformly in scale and in ξ.*

Proof. The load-bearing hypotheses of Naud's theorem for G_a — the branch forms L_k of (3), the endpoints ρ_k, the expansion $\inf |(G^2)'| = 3$, and the UNI non-degeneracy (slopes $-\rho_k$ pairwise distinct, Proposition 2) — are *exact algebraic facts of the fixed integer matrices P_k*, with no dependence on a. Only the $O(1/a)$ translation corrections carry scale, and they are lower order, perturbing the roof's constants but not the non-degeneracy. Hence $\{G_a\}_{a \ge a_0}$ is a uniformly-controlled family of real-analytic expanding UNI maps, and Naud's constants $(C, \theta, \varepsilon_0)$ may be taken uniform over it. Applying the theorem to G_a at frequency $\xi \sim a$ — never perturbing from G — gives the gap at native resolution. □

Remark 2. The gap's scale-uniformity is confirmed directly: the twisted $|\lambda(i\xi)|$ computed on states restricted to a bit-length band is flat across bands and across native-resolution frequencies $\xi \in \{2, 5, 12, 20, 35\}$, and the smallest-scale band — the largest correction — is if anything more contractive.

7 The joint local limit theorem and the hazard bound

By the Nagaev–Guivarc'h method [8, 9, 10], and in the Gibbs–Markov form of Aaronson–Denker [11], the operator $\mathcal{L}_{i\xi}$ is analytic in ξ with leading eigenvalue $\lambda(i\xi)$, $\lambda(0) = 1$, $P(i\xi) = \log \lambda(i\xi)$ satisfying $P(0) = 0$, $P'(0) = i\Lambda$, $P''(0) = -\sigma^2$ with $\sigma^2 \approx 0.265 > 0$ and non-lattice roof (both verified). The gap of Proposition 5 makes the twisted resolvent integrable, and this is all the density-zero conclusion needs.

Proposition 6 (Native-resolution hazard, upper bound)**.** *Under the scale-uniform gap, the joint density $p_d(z, \ell)$ of (z_d, ℓ_d) is bounded uniformly in d down to scale $1/S_d$ in z; hence at a crossing of scale S,*

$$\Pr[o_d = m \mid \text{scale} = S] \le \frac{C}{S} \ (m \in \{1, 3, 5\}), \qquad \Pr[o_d \in \{1, 3, 5\} \mid S] \le \frac{C'}{S}.$$

Proof. $\mathbb{E}[\mathbf{e}(\xi\ell_d)\psi(z_d)] = \lambda(i\xi)^d\Pi_\xi(\psi) + O(\vartheta^d)$ with $\vartheta < 1$ from the gap. Fourier inversion in ξ gives p_d, and the gap makes the inversion integral converge with $\sup_{z,\ell} p_d \le C$ uniformly in d — the standard local-limit *upper* bound, using only integrability of the twisted resolvent, not the CLT expansion or a precise constant. Since $o_d = m \iff z_d = m/S$ and one integer offset occupies a z-window of width $\asymp 1/S$, integrating p_d over that window gives $\Pr[o_d = m] \le C/S$. □

Remark 3 (What is and is not enough). The macroscopic bound $h \in L^\infty$ (Rychlik) controls z_d at fixed resolution but *not* at scale $1/S$: integer orbits could a priori concentrate there. Ruling that out uniformly in d is exactly the content of the gap, which is why the additive-roof route is load-bearing and not a formality. Directly on integer orbits the bound is confirmed: over a 2-D seed ensemble, $S \cdot \Pr[o_d \in \{1,3,5\}]$ is flat at ≈ 3–4 across the scale window where halting occurs, consistent with $C' = 6h(0)$, $h(0) \approx 0.5$.

8 The count

Theorem 1 (Density form)**.** *Under Naud's Theorem 2.3 and the Nagaev–Guivarc'h LLT,* $\lim_{N\to\infty} N^{-2}\#\{(a,b) : a,b \le N,\ \mathbf{R}\text{-orbit meets } \mathcal{D}\} = 0$*, and* $\#\{\text{halts} \le N\} = O(N \log N)$*.*

Proof. Scales grow geometrically, $S_d \asymp s\lambda^d$ for a seed of size s (Lemma on $\Lambda > 0$, companion paper). By Proposition 6 the size-s halting fraction is $\mathrm{hf}(s) = \sum_{d\ge 0} \Pr[\text{fatal at } d \mid s] \le \sum_d C'/(s\lambda^d) \le C''/s$. Seeds equidistribute in the initial coordinate with discrepancy $O(1/a)$ (lattice-point counting in sectors) and mix to μ, so with $\#\{a \in [s,2s], b \le N\} \asymp sN$,

$$\#\{\text{halts} \le N\} \le \sum_{s \text{ dyadic}} \mathrm{hf}(s) \cdot sN \le C'' \sum_{s \le N} \frac{1}{s} \cdot sN = O(N \log N) = o(N^2).$$

The prediction is confirmed: $\#/(N \log N) \in [3.06, 3.61]$ for $N = 10^3$–6.4×10^4. □

9 Status: proved, cited, and to be written

Proved here (or in the companion paper), from the arithmetic: the reduction to G; endpoints, expansion, summable variation; the ACIM and mixing (Rychlik, Hofbauer–Keller); transience and $\Lambda > 0$; the first-fatal decomposition (1); the suspension and the model roof (4); UNI for τ via non-proportional branch forms (Proposition 2 = Naud's NLI); the exact geometric tail rate $\lambda^{-1} = 1 - \rho^*$ (Proposition 4); the scale-independence of the Dolgopyat hypotheses (Proposition 5); $\sigma^2 > 0$, non-lattice; the count (Theorem 1). The scale-uniform and countable-branch gaps and the hazard bound are additionally confirmed numerically on the exact integer dynamics.

Cited, with hypotheses discharged: the twisted spectral gap (Naud [6], Theorem 2.3; UNI verified), and the Nagaev–Guivarc'h local limit theorem ([10, 11]; $\sigma^2 > 0$, non-lattice verified). We invoke only the *upper* half of the LLT (Proposition 6), the easy consequence of the gap.

To be written in full (standard, no new idea): (a) the sequential-composition argument that the uniform per-scale gap (Proposition 5) yields the orbit's native-resolution density bound, together with the integer upper bound it feeds; (b) the propagation of UNI from adjacent branches (Proposition 2) to all word-lengths, standard for real-analytic maps; (c) the detailed Nagaev–Guivarc'h perturbation for the joint operator and the seed-equidistribution discrepancy carried through the sum.

What is not claimed. The density form is not asserted as an unconditional theorem: it rests on the two cited results above, whose hypotheses we have verified but whose full application we have

not written out. The measure form is unconditional. The specific certified orbit $(6, 13)$ is μ-generic and deciding its own fate is Collatz-hard; nothing here bears on it. This is the same three-tier posture — unconditional generic result, conditional sharp result, untouched pointwise result — that Tao's theorem [1] occupies relative to the Collatz conjecture.

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